Minjae Koo

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EDUCATION

The University of Michigan — **Ann Arbor, MI**

M.S. in Electrical & Computer Engineering — *Graduation: May 2028*

B.S. in Computer Engineering | GPA: 3.86 — *Graduation: Dec 2026*

- **Relevant Coursework:** VLSI Chip Tapeout Lab, VLSI Design I, Computer Architecture, Digital Integrated Circuits, Analog Circuits, Logic Design, Electronic Circuits, Signals and Systems, Programming/Data Structures
- **Awards:** Dean's Honor List (x5 sem), 2025-26 Ernest W. Reynolds Endowed Scholarship in EE, James B. Angell Scholar

PROJECTS

Compute-In-Memory ASIC Tapeout - *Cadence Virtuoso, Innovus, TSMC 65nm, Chip Tapeout* — *Jan 2026 - Apr 2026*

- Directed the end-to-end TSMC 65nm tapeout of an all-digital Compute-In-Memory ASIC as Team Captain (5 members), taking ownership of verification, synthesis, physical design, top-level integration, and physical signoff
- Led and verified a mid-project architecture pivot, transitioning from a hand-routed 6T SRAM core to a synthesizable latch-based RTL core to resolve peripheral routing issues without delaying tapeout schedule
- Developed the complete frontend and test infrastructure using SystemVerilog and Synopsys Design Compiler

2-way Superscalar OoO Processor - *SystemVerilog, Synopsys Design Compiler* — *Mar 2026 - Apr 2026*

- Collaborated in a 5-member team to upgrade a 3-stage LC2K processor into a 2-wide superscalar in SystemVerilog; implemented an 8-entry Microop Queue to allow limited OoO dispatch while handling hazards and forwarding
- Developed the front-end with a direct-mapped I-cache, 2-banked BTB, and 2-bit branch predictor; improved execution throughput by breaking load/store instructions into internal micro-ops and routing them to specific execution lanes
- Verified processor functionality using Verdi waveforms and synthesized the design to evaluate hardware tradeoffs

16-bit RISC-V Processor Layout - *Cadence Virtuoso, Innovus, TSMC 65nm* — *Sep 2025 - Dec 2025*

- Collaborated in a team of 5 to design and layout a custom 2-stage pipelined 16-bit RISC-V processor in TSMC 65nm
- Designed and hand-laid out custom schematics for the 16-bit register file, ALU, logarithmic shifter, and a 16x16 8T-SRAM; successfully integrated manual layouts with an APR-generated program counter and instruction decoder.
- Verified processor functionality by running a Fibonacci benchmark through NCVerilog testbenches and pre/post-layout SPICE simulations; achieved a fully DRC/LVS-clean design with a 162 µm × 200 µm core area operating at 200 MHz

RELEVANT EXPERIENCE

SK Hynix Memory Solutions America — **San Jose, CA**

Incoming SoC Physical Design Intern — *Sep 2026 - Mar 2027*

Nissan Motor Corporation — **Farmington Hills, MI**

R&D Engineering Intern, ZEV Team — *May 2026 - Aug 2026*

- Designed and developed a 16-channel relay driver PCB for Vehicle-to-Load (V2L) test automation, driven by an STM32F103 microcontroller to support both automated and manual high-current multi-channel switching sequences
- Conducted DC chargeability testing for the 2026 Nissan LEAF across varying temperatures and battery states of charge
- Benchmarked battery preconditioning strategies across competitor EV to identify industry trends

LG Energy Solution — **Holland, MI**

Manufacturing Intern, Cell Equipment Technology Team — *May 2025 - Aug 2025*

- Designed, developed, and implemented an HMI-based PET film roll reuse tracking system in MELSOFT GT Works3 using QR codes integrated with MES, enabling reuse rate tracking and reducing downtime by 7 hrs/month
- Modified motor cover by adding window cutout designs in SolidWorks, reducing operator PM time by ~70%
- Acquired operational understanding of the Lithium-ion battery cell assembly process and systems

Supermileage Project Team — **Ann Arbor, MI**

Power Electronic Subteam Member — *Oct 2023 - Dec 2024*

- Designed a 2-layer PCB relay driver board with MOSFET low-side switching for E-switch application
- Mentored ~10 new members in soldering techniques, oscilloscope use, and PCB design workflow during onboarding

SKILLS

- **Languages:** SystemVerilog, Verilog, Python, C/C++, TCL
- **EDA & Design Tools:** Cadence Virtuoso, Innovus, Synopsys DC, Verdi, KiCad, LTSpice, ModelSim, AI tools